

AI/ML Task 2 Report

Feature Engineering, Model Optimization & Performance Comparison

Maincrafts Technology — AI/ML Internship

1. Objective

Building on Task 1's basic Linear Regression model, this task implements an enhanced House Price Prediction system: scaling input features, training and comparing multiple regression algorithms, and selecting the best-performing model using objective evaluation metrics (RMSE and R^2 Score).

2. Dataset

The task specifies the scikit-learn California Housing dataset (target: Median House Value; features: Median Income, House Age, Average Rooms, Population, and location attributes). Live download of this dataset requires external network access that was not available in the execution environment used for this report, so a synthetic dataset (20,640 rows) was generated that mirrors the real dataset's feature names, numeric ranges, and relationships between features and price. The notebook includes the exact `fetch_california_housing()` call, commented out, ready to run unmodified on Google Colab or any machine with normal internet access.

3. Methodology

Feature scaling: All 8 input features were standardized using `StandardScaler` (zero mean, unit variance) to ensure fair learning across features of very different magnitudes (e.g. Population vs. Average Occupancy).

Train-test split: 80% training / 20% testing, `random_state=42`, so performance is measured on unseen data.

Models trained: Linear Regression (baseline), Ridge Regression (`alpha=1.0`, L2-regularized to reduce overfitting), and Decision Tree Regressor (`max_depth=5`, to capture non-linear relationships while limiting depth to control overfitting).

4. Results — Model Performance Comparison

Model	RMSE	R^2 Score
Linear Regression	0.4934	0.7542
Ridge Regression	0.4934	0.7542
Decision Tree (<code>max_depth=5</code>)	0.4968	0.7509

Lower RMSE = better prediction accuracy. Higher R^2 = better explanatory power. Linear Regression achieved the lowest RMSE (0.4934) and the highest R^2 score (0.7542), very marginally ahead of Ridge Regression, with the Decision Tree trailing slightly on both metrics.

5. Visual Performance Validation



Actual vs. Predicted House Prices for the best model (Linear Regression). Points cluster closely around the red 45° reference line, indicating consistent, unbiased predictions across the price range, with slightly more spread at the highest price values.

6. Model Selection & Justification

Linear Regression is selected as the final model. It produced the lowest RMSE and highest R^2 among the three candidates, and the underlying feature-target relationships in this dataset are largely linear, so a simple linear model fits at least as well as a regularized (Ridge) or non-linear (Decision Tree) alternative — while remaining the most interpretable and computationally cheapest option. Ridge Regression's near-identical results confirm that overfitting is not a significant issue here, so the added regularization provides negligible benefit. The Decision Tree's marginally worse performance suggests that, for this data, splitting on individual feature thresholds captures the linear trend less precisely than a global linear fit.

7. Conclusion

This task demonstrated a professional ML workflow beyond basic model training: correct feature scaling, a proper train/test split, training multiple algorithms, comparing them with objective metrics, and justifying a final model choice with evidence rather than assumption — mirroring how ML engineers iterate on models in real-world projects.